

STILLWATER, OK, USA

WMO: 723545

Lat: 36.162N

Lon: 97.089W

Elev: 300

StdP: 97.77

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 03965

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-7.9	-16.9	0.9	-7.6	-14.1	1.1	-5.2	11.8	10.4	10.9	9.9	3.2	0	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	38.9	23.7	37.3	23.9	35.3	24.1	26.0	34.1	25.5	33.4	24.9	32.6	5.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.4	30.1	23.0	18.4	29.0	22.6	18.0	28.6	82.3	34.1	79.6	33.3	77.0	32.7	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.8	8.8	-15.6	40.6	3.4	2.4	-18.1	42.4	-20.1	43.8	-22.0	45.1	-24.4	46.8		
			-15.9	27.4	3.3	1.3	-18.2	28.3	-20.2	29.1	-22.0	29.9	-24.4	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	3.5	5.6	10.7	15.7	20.5	25.5	28.3	27.7	23.4	16.5	10.1	4.4
	DBStd	10.01	5.84	6.03	6.14	5.16	4.29	3.14	3.02	3.19	4.14	5.11	5.36	5.57
	HDD10.0	694	213	146	65	12	1	0	0	0	0	8	64	186
	HDD18.3	1987	459	358	245	109	28	1	0	0	7	96	252	432
	CDD10.0	2905	12	23	87	183	327	464	567	550	403	210	66	13
	CDD18.3	1155	0	1	9	30	96	215	309	292	160	40	4	0
	CDH23.3	13382	3	11	91	326	932	2407	3919	3536	1708	406	43	1
	CDH26.7	6698	0	2	18	91	328	1134	2221	1969	801	128	6	0
Wind	WSAvg	4.2	4.1	4.3	4.9	5.1	4.5	4.4	3.8	3.6	3.7	4.0	4.1	4.0
Precipitation	PrecAvg	879	31	36	69	78	124	109	76	78	103	76	58	40
	PrecMax	1179	93	137	196	211	371	268	209	216	315	222	169	130
	PrecMin	500	0	2	0	4	15	21	0	1	1	10	0	2
	PrecStd	171	27	31	47	49	72	60	49	54	66	53	47	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.1	24.8	28.8	32.3	33.9	37.8	41.3	42.0	38.9	32.4	27.3	22.3
		MCWB	12.1	14.9	17.1	19.4	22.0	24.0	23.3	22.9	23.3	21.3	17.0	14.2
	2%	DB	19.1	21.9	26.0	29.0	32.2	35.8	39.0	38.9	34.9	30.0	24.1	18.8
		MCWB	10.7	13.0	15.6	18.1	22.1	23.8	23.6	23.9	23.0	19.9	15.4	13.1
	5%	DB	16.3	18.7	23.0	27.2	30.2	33.8	37.3	37.3	33.0	27.7	22.0	16.2
		MCWB	9.5	12.3	14.8	17.7	21.3	23.9	24.2	23.9	22.7	18.7	14.7	10.4
	10%	DB	13.0	16.1	21.0	24.8	28.1	32.4	35.4	35.1	31.4	25.4	19.0	13.6
		MCWB	7.6	10.4	13.8	17.0	20.8	23.5	24.4	23.9	21.9	17.8	13.2	9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.3	16.9	19.2	21.6	24.6	26.8	27.0	26.7	25.2	23.1	19.3	17.2
		MCDB	18.7	21.2	25.6	28.2	30.5	33.3	34.4	34.9	33.1	29.1	23.8	20.3
	2%	WB	12.4	15.0	17.8	20.2	23.5	25.7	26.1	25.7	24.2	21.9	17.4	14.0
		MCDB	17.1	19.5	23.3	26.2	29.4	32.7	34.1	34.5	31.9	27.1	21.9	16.8
	5%	WB	10.0	13.0	16.5	19.4	22.7	25.0	25.5	25.1	23.6	20.7	15.8	11.4
		MCDB	15.6	17.9	21.5	24.9	28.5	31.7	33.5	33.7	30.8	25.3	20.0	14.9
	10%	WB	7.9	10.3	14.8	18.2	21.7	24.2	25.0	24.5	22.9	19.4	14.3	8.9
		MCDB	12.5	15.7	19.8	23.4	27.1	30.4	32.8	32.4	29.4	23.7	18.2	13.3
Mean Daily Temperature Range		MDBR	12.6	12.6	12.9	12.9	11.9	11.5	12.3	12.7	13.0	13.3	13.0	11.9
	5% DB	MCDBR	17.7	17.0	15.8	15.4	13.5	13.2	14.5	15.2	14.4	15.5	16.1	16.1
		MCWBR	10.5	10.0	8.4	7.9	6.0	5.0	4.3	4.4	5.4	7.3	8.8	10.3
	5% WB	MCDBR	15.9	14.7	12.3	12.9	11.4	11.1	12.2	12.5	12.1	11.6	13.1	13.3
		MCWBR	10.3	10.1	7.3	7.8	6.0	5.2	4.8	4.8	5.0	6.6	8.6	10.1
Clear-Sky Solar Irradiance	taub	0.295	0.301	0.340	0.380	0.417	0.427	0.432	0.433	0.391	0.346	0.312	0.289	
	taud	2.498	2.495	2.414	2.319	2.288	2.294	2.320	2.258	2.361	2.473	2.507	2.526	
	Ebn at Noon	908	944	928	903	870	857	850	842	863	878	880	893	
	Edh at Noon	81	92	109	127	133	132	128	133	114	93	80	74	
All-Sky Solar Radiation	RadAvg	2.68	3.31	4.32	5.37	5.90	6.74	6.74	6.10	5.07	3.82	2.84	2.26	
	RadStd	0.27	0.49	0.28	0.37	0.59	0.59	0.37	0.37	0.42	0.47	0.35	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-1.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page